

Engineering Approaches for Breast Cancer Diagnosis: A Review

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(Methodological Review)

Abstract—Breast cancer is a leading cause of mortality among women. The patient's survival rate is uncertain due to the limitations in the accuracy of diagnosis and effective monitoring during cancer treatment. The key to efficaciously controlling cancer on a larger scale is effective diagnosis at an early stage of cancer by distinguishing the vital signatures of the diseased from the normal breast tissue. The breast tissue is a heterogeneous turbid media that exhibits multifaceted bulk tissue properties. Various sensing modalities can yield distinct tissue behavior for cancer and adjacent normal tissues, serving as a basis for cancer diagnosis. A novel multimodal diagnostic tool that can concurrently assess the optical, electrical, and mechanical bulk tissue properties can substantially augment the clinical findings such as histopathology, potentially aiding the clinician to establish an accurate and rapid diagnosis of cancer. This review aims to discuss the clinical and engineering aspects along with the unmet challenges of these physical sensing modalities, primarily in the field of optical, electrical, and mechanical. The challenges of combining two or more of these sensing modalities that can significantly enhance the effectiveness of the clinical diagnostic tools are further investigated.

Index Terms—Breast cancer, biopsy needle, near-infrared spectroscopy, polarization spectroscopy, bioimpedance, MEMS.

I. INTRODUCTION

BREAST cancer is one of the leading causes of cancer death worldwide, as observed from the latest report by the International Agency for Research on Cancer (IARC) in 2020

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[1]. It is the most commonly diagnosed cancer, accounting for 11.7 percent of all cancer cases, followed by lung cancer [1]. The most common protocol for diagnosing and treating breast cancer involves identifying the early symptoms by physical self-examination, followed by a visit to the physician for Clinical Breast Examination (CBE). The early-stage screening, apart from physical examination by the clinician, involves imaging with X-ray mammography [2], sonography [3], Magnetic Resonance Imaging (MRI) [4], and blood chemistry studies such as complete blood count (CBC). Through contrast and density abnormalities in the acquired images, these imaging techniques aid in confirming the presence of a lesion within the breast. X-ray mammography is the commonly used tool for breast cancer screening that measures the attenuation of X-rays while propagating through the breast tissue. Mammography significantly reduced the mortality through early detection of cancer for women over 50 years, with a sensitivity and specificity of 77% and 97%, respectively [5]. However, the sensitivity and specificity of X-ray mammography are reduced to 67% and 89%, respectively, for women with dense breasts [6]. Furthermore, an X-Ray mammography scan poses the risk of radiation while yielding lower accuracy for patients of age less than 50 [7], [8]. MRI involves exciting and aligning the hydrogen atoms (i.e., protons) with the help of powerful magnets and tells the properties of the tissue based on the alignment and energy released by the protons after turning off the radiofrequency current. The sensitivity of Contrast-enhanced MRI is reported between 93% and 100% [9], [10], albeit with lower specificity. Moreover, MRI is bulky and time-consuming. MRI is not generally considered as a modality for early-stage screening due to its high cost.

The ultrasound technique exploits acoustic impedance mismatch between the types of tissue as high-frequency acoustic waves propagate through the breast. Breast ultrasonography is a highly sensitive (81%), specific (88%), non-ionizing, and low-cost technology for the detection of breast cancer in symptomatic women [11]. Furthermore, ultrasonography is a suitable choice for early detection of breast cancer in younger (under 30) or pregnant women [12], but it requires skilled personnel, and reproducibility is low [13] compared to mammography. Another technique involves a cytology procedure called fine-needle aspiration cytology (FNAC) to confirm whether the

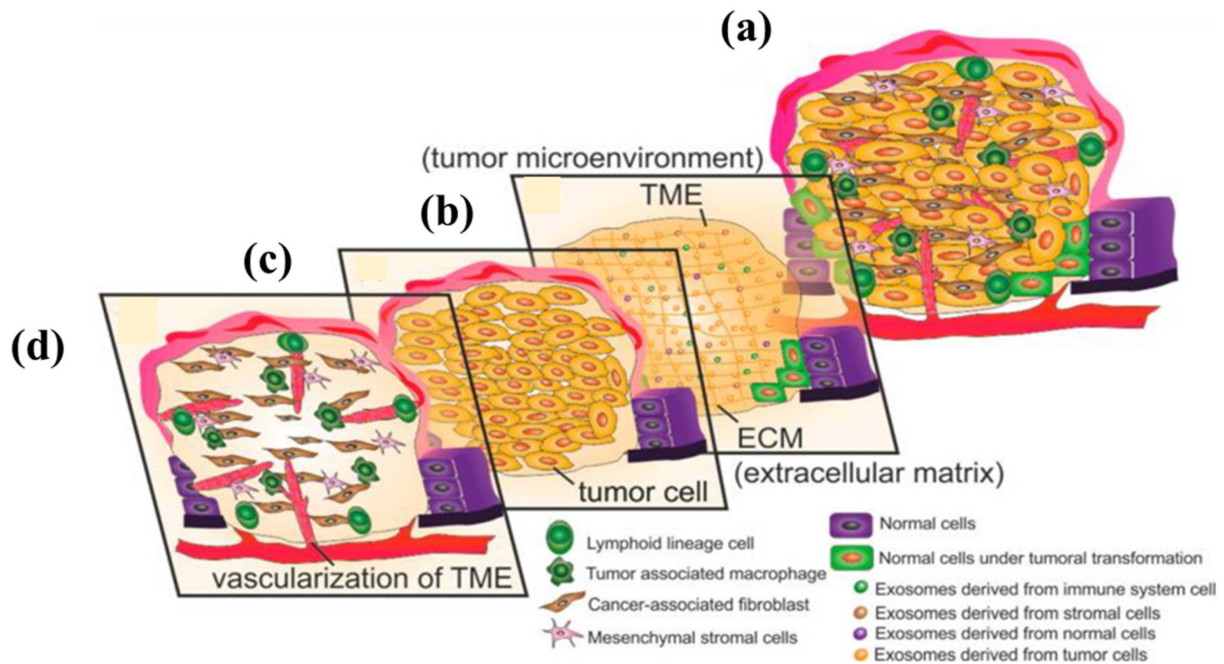


Fig. 1. Structure of the Tumour Microenvironment (TME). (a) Complex and highly heterogeneous TME of a late stage tumor. (b) Exosomes are present in the conversion of non-cancerous cells into cancerous cells contiguous to the TME and play vital role in the autocrine and paracrine communication amid TME cells. In the TME, ECM is usually very dense and rigid, drive desmoplasia. (c) The TME is very aggressive, lack of nutrients and oxygen including the Warburg effect and drives in acidification of the TME. (d) Angiogenesis which is produced due to fast growth of cancerous cells, Immune system's cells and stromal cells plays a vital role in the development and prediction of tumor [28]., with the permission of MDPI, copyright 2019.

tumor is malignant or benign. In FNAC, biopsy removes a small cluster of cells from the suspected tumor region, often guided by ultrasound. The cells thus extracted are examined for their altered morphology through hematoxylin and eosin (H&E) staining to confirm malignancy. Another minimally invasive type of biopsy, known as a core needle biopsy, may also be performed to ascertain the malignancy's biochemical nature (primarily ER, PR, and HER2 status) [14]. Based on the results from these biopsy procedures, a personalized treatment plan is chalked out that could include lumpectomy/mastectomy with or without chemotherapy and radiation treatment depending upon the staging and type of cancer [15].

Although these techniques have become standard practice, there are some inherent limitations. The FNAC biopsy procedure is usually performed under the intraoperative guidance of ultrasound or preoperative guidance via computed tomography (CT) or MRI. However, in some patients where there is an increased density of breast tissues or a microscopic-sized lesion, it becomes challenging to image the lesions and distinguish them from the normal tissues, making it difficult to sample from the targeted location [16], as well as highly dependent on the personnel skill performing the biopsy procedure. Recently developed non-conventional techniques are being studied in order close the gap in clinical utility like elastography [17], computer-aided tomosynthesis (3-D mammogram) [18], [19], electromagnetic breast imaging [20], photoacoustic spectroscopy [21], electrical impedance spectroscopy [22], optical spectroscopy [23], and micro-electromechanical systems (MEMS)-based mechanical sensing technologies for the diagnosis of breast cancer.

In the broader perspective, to overcome breast cancer, a translational multimodal system incorporating various sensing modalities involving optical, electrical, and mechanical elements needs to be thoroughly tested and validated. This new information can potentially aid the clinician, in addition to histopathology, in making an accurate breast cancer diagnosis. The potential increases in global accessibility to such technologies and the latest advancements in miniaturizing the modalities will enable the prospective tool to become portable, affordable, and easy to operate for clinicians. Moreover, this review discusses the fundamentals, applicability in clinical settings, and unmet challenges of these novel physical sensing modalities for effective breast cancer diagnosis.

II. TUMOR MICROENVIRONMENT

The tumor microenvironment (TME) includes the immune cells (e.g., Macrophages, T and B lymphocytes, etc.), blood vessels, extracellular matrix (ECM), and fibroblasts, as shown in Fig. 1. The tumor and its neighboring microenvironment are highly interdependent [24]. The partial pressure of O_2 in cancerous tissue is usually low compared to normal tissue. Therefore, TME is generally characterized by regions of O_2 depletion, viz. hypoxia, and anoxia. For the proper functioning of natural tissue, the concentration of O_2 should lie in the range of 2% to 9%. When the O_2 concentration decreases below 2%, the associated tumor is defined as a hypoxic region, and the state of complete deprivation of O_2 is known as anoxia [25]. TME is favorable for the high proliferation of cancer cells, associated with increased

glucose metabolism and decreased pH, commonly referred to as the Warburg effect [26]. As a result, the cancerous tissue's morphology is altered from normal tissue, wherein the former is heterogeneous while the latter is homogeneous in structure [27]. The different stages of transformation of the normal tissue to cancerous tissue include hyperplasia, dysplasia, and finally, neoplasia. These changes demand an altered energy budget in the TME, which is the underlying reason for the biochemical changes [28]. These reactions are managed by stromal cell signaling (e.g., autocrine and paracrine) close to the tumor and provide information about the possibility of tumor metastasis [29].

III. RECENT ADVANCES IN NON-INVASIVE TECHNOLOGIES FOR BREAST CANCER DIAGNOSIS

Sensing modalities such as optical, electrical, and mechanical can play a vital role in understanding various aspects of tissue properties. The sensing modalities for breast cancer diagnosis can be categorized based on contact or non-contact techniques. Point-based optical diagnosis can help diagnose the breast tissue due to changes in optical properties of the tissue caused by malignancy. Similarly, non-conventional electrical and mechanical technologies can also help understand the state of the breast tissue and provide novel biomarkers that can help the clinician to make a rapid decision for the effective diagnosis of cancer with high accuracy. The bulk tissue characteristic reorganizes as the tissue undergoes a transition from normal to malignancy; these disparities can be accurately quantified by the individual as well as hybrid sensing techniques, as discussed in the following sections of this review.

A. Diffuse Optical Tomography (DOT)

DOT is a non-invasive, non-ionizing, and highly sensitive technology that measures the functional properties of breast tissue by exploiting the near-infrared (NIR) spectrum (600–1100 nm) of light [30]. The source and detector configuration in a DOT system can be either a pentagonal or football-shaped interface. In this configuration, the breast tissue is slightly compressed at the tissue-optical fiber interface by an adjustable football-shaped housing integrated with source and detector fibers [31], primarily focused on 2D imaging of the breast tissue. In contrast, multilayer ring geometry contains a source and detector combination in each layer. The system comprises 32 fibers (i.e., four fibers in each layer) dedicated for top surface imaging, while 32 fibers are dedicated for bottom surface imaging of the breast, respectively [32] is helpful for 3D imaging by moving the source-detector ring up and down. Other configurations like parallel plane [33], handheld structures [34], and wearable structures [35] have also been used in DOT imaging of breast tissue. Zhao *et al.* [31] developed a portable multiwavelength frequency-domain (FD) (660–850 nm) and continuous-wave (CW) (850–1064 nm) technique-based NIRST system for breast cancer detection. They used laser diodes and photomultiplier tube detectors with a novel gain adjustment technique to reconstruct the tomographic collagen content images in the nine healthy and one cancer subjects ($N = 10$). Flexman *et al.* [32]

proposed a CW technique-based system to study the hemodynamic effects of the breast tissue. The system used a multiwavelength (765, 805, 827, 905 nm) NIR light source and acquired the data from ($N = 21$) subjects containing three healthy and 18 diseased subjects using 32 light sources and 64 detectors. Wang *et al.* [33] have developed a compact fiber-free CW technique-based multiwavelength system to acquire the three-dimensional optical properties of the breast tissue from one subject ($N = 1$). The system incorporates 56 multiwavelength (660, 750, and 840 nm) LEDs and 56 photodiodes with high gain, low-noise current amplifiers to detect breast cancer and perform neoadjuvant chemotherapy (NAC) efficacy monitoring. Spink *et al.* [35] proposed a wearable CW technique-based probe to study the breathing hemodynamics behavior in breast tissue and monitor the NAC response in ($N = 4$) breast cancer subjects. The probe incorporates 16 dual-wavelength (750 and 850 nm) LEDs and 16 detectors with an in-built trans-impedance amplifier, as shown in Fig. 2(a). The developed probe is encased in biocompatible housing and conformal to the curved surface of the breast, as shown in Fig. 2(b) and the obtained results in Fig. 2(c). Shokoufi *et al.* [36], [37] reported a compact handheld optical scanner probe using multiwavelength LEDs to detect breast cancer. They reported an affordable system different from fiber-optic based illumination using four different wavelengths (e.g., 690 nm, 750 nm, 800 nm, and 850 nm) LEDs in a single package. They reconstructed the images of breast tissue phantom at different positions by using a charge-coupled device (CCD) array detector.

B. Electrical Impedance Spectroscopy

Electrical impedance sensing will be a promising technology for point of care diagnosis of cancer treatment that results in effective and continuous monitoring of the efficacy of drugs in real-time, which ultimately help in reducing the associated costs and logistics in the cancer treatment and improves survival rates and reduce the trauma. In this direction, Abramson *et al.* [38] developed a flexible electronic strain sensor for monitoring the tumor regression *in-vivo* in real-time. They reported flexible, autonomous sensors tumor (FAST) wireless technology-based sensor for tumor volume regression measurement in a rodent model. The developed sensor measures the change in resistance as the tumor progress or regresses. This change in resistance is related exponentially as strain grows due to tumor progression. The flexible electrical sensor can also be used for other types of cancer occurring in the abdomen [39] and skin [40], [41]. The wireless flexible electrical sensor will change the approach to treating cancer in the forthcoming years. Another work by Dixon *et al.* [22] reported a handheld, battery-operated imaging device named Clear Edge (CE) to determine the tumor margin of excised breast tissue intraoperatively from ($N = 112$) subjects with sensitivity (84.3%) and specificity of (81.9%). The operating principle of the said device is bio-impedance spectroscopy, which is extremely sensitive to changes in the tissue's dielectric characteristics due to malignancy and has a 3 mm penetration depth. CE has an LCD with greens (fatty normal), yellows (fibrous normal), and reds (cancerous) color displayed in each pixel based on the malignancy.

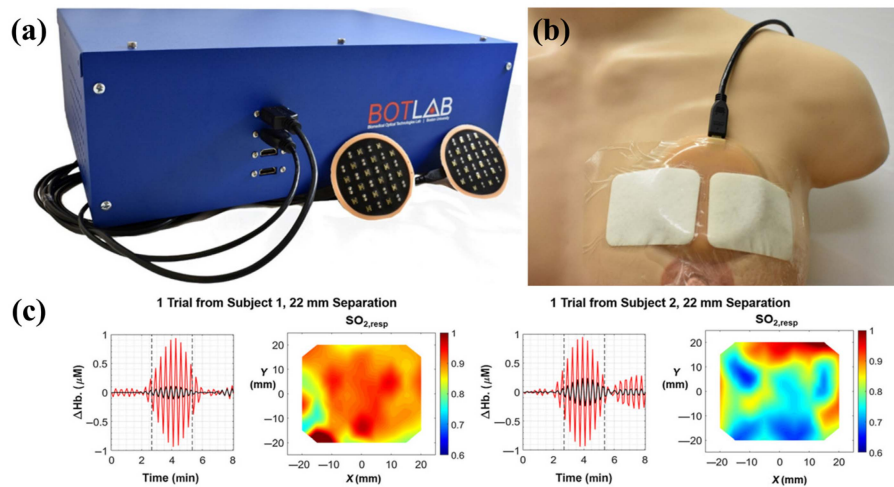


Fig. 2. Illustration of a wearable optical spectroscopy-based device. (a) Actual images of handheld encased probe connected with control system. (b) Probe attached to the human bust model performing the scanning of the breast. (c) Mean time traces of change in deoxyhemoglobin (black), oxy-hemoglobin (red), and topographic map of SO_2 for subject 1 and subject 2, respectively [35], with the permission of SPIE, copyright 2021.

C. Ultrasound Sensing Using PMUTs

Piezoelectric Micromachined Ultrasonic Transducers (pMUTs) is a new generation of devices developed by MEMS-based manufacturing techniques and employ tiny membranes or films driven by either piezoelectric or capacitive actuation (pMUTs and CMUTs, respectively). Thickness-mode pMUTs are suitable for high-frequency medical imaging (e.g., intravascular imaging). Whereas flexural vibration-mode pMUTs are well suited for low-frequency imaging [42].

Conventional micromachining technologies are used to fabricate Thin-film piezoelectric MEMS devices. Ion beam sputtering (IBS), RF planar magnetron sputtering or DC magnetron sputtering are examples of physical approaches. Metal-organic-chemical vapor deposition (MOCVD) and chemical solution deposition (CSD) with their various paths involving sol-gel processes, or metal-organic-decomposition (MOD), pulsed-laser deposition (PLD) are now well-proven techniques for PZT deposition [43]. The deposition of high-quality piezoelectric films typically necessitates using buffer layers to avoid inter-diffusion and oxidation and reduce residual tensions and hence limit piezoelectric properties deterioration. Sacrificial releasing, front-side etching, cavity silicon-on-insulator (SOI) wafer bonding, and backside etching are the four methods most typically employed to create the membrane [44]. Mechanical energy is converted to electrical energy or vice versa via electromechanical transduction, enabled by piezoelectric materials. This one-of-a-kind electromechanical transduction effect has given rise to new sensor, actuator, and transducer applications. An ideal ultrasound device must have a high coupling coefficient and thermal stability, excellent acoustic impedance similar to tissue impedance, low mechanical and electrical losses, and a large dielectric constant, all of which contribute to a highly sensitive transducer for a broad bandwidth of operation [45]. Smaller piezoelectric devices provide advantages such as

decreased noise levels and power consumption, making them indispensable for integrated systems, multi-functionality, and batch production [46].

Breast fibro glandular tissues have a higher speed of sound than fatty tissues. Compared to normal tissues, cancerous breast tissue has a higher speed of sound because of the increased density of cells in cancer. According to the reported work by Nebeker *et al.* [47], the average sound speed for fatty tissue is 1478 m/s, 1510 m/s for fibro glandular breast tissue, 1513 m/s for benign breast tissue, and 1548 m/s for malignant breast tissue, respectively. Similarly, the cancerous tissue has higher acoustic attenuation (i.e., ultrasonic absorption) than adjacent normal tissues due to the increased proliferation of fibrous and connective tissue around the tumor [48]. Therefore, multimodal techniques can help improve the sensitivity and accuracy in delineating cancerous from adjacent normal tissues. Ruby *et al.* [49] reported a novel ultrasound technique to quantify the change in speed of sound (ΔSoS) by using a traditional ultrasound system to differentiate between benign and malignant solid lesions from ($N = 20$) subjects. Ten subjects with fibroadenoma and ten subjects with histologically confirmed breast cancer. The peak (ΔSoS) in breast cancer subjects is significantly high as compared to fibroadenoma subjects with sensitivity (90%) and specificity (80%). pMUTs based devices can be a promising tool to quantify the acoustic properties and perform the imaging of the diseased region of the tissues [42].

IV. RECENT ADVANCES IN MINIMALLY INVASIVE TECHNOLOGIES FOR BREAST CANCER DIAGNOSIS

Minimally invasive technologies that use multimodal (i.e., optical, electrical, and mechanical) sensing modalities can help clinicians better understand various characteristics of breast tissue, resulting in a more accurate diagnosis of breast cancer. The surface and bulk tissue characteristics reorganize as tissue

transitions from normal to malignancy; individual and multi-modal sensing can detect these disparities effectively to aid in the diagnosis of breast cancer.

A. Optical Sensing

The optical sensing technique, which uses visible NIR light to quantify the bulk optical characteristics of tissue minimally invasively, is a promising technology. Various optical sensing techniques are used for breast cancer diagnoses, such as Diffuse Reflectance Spectroscopy (DRS), Fluorescence Spectroscopy (FS), Raman Spectroscopy (RS), Photoacoustic Spectroscopy (PAS), and Polarization-Sensitive Spectroscopy (PSS). This section discusses optical spectroscopy-based devices that can potentially be translated into portable diagnostic tools for breast cancer diagnosis. Optical spectroscopy-based techniques have been explored widely as tools for rapid breast cancer diagnosis as this technique is fast, non-ionizing, and minimally invasive [50]–[52]. The variations in optical properties of cancerous and normal tissue have been extensively studied [52], with the potential for using these variations in optical properties for breast cancer diagnosis. Cancerous tissue undergoes higher absorption of light due to increased hemoglobin content [53], [54], reflecting tumor angiogenesis and higher scattering of light because of increased collagen content, changes in the tissue microstructure, high anisotropy, and concentrations of cell organelles [55], forming a basis of differentiating between normal and cancerous tissues. Gharthey *et al.* [56] developed a portable, battery-operated screening tool (Breast-i) based on a transillumination method for breast cancer detection at an early stage with a sensitivity of 93%. However, Edge *et al.* [57] was critical of the transillumination-based screening and suggested extensive clinical trials before accepting transillumination-based tools as an alternative modality for breast cancer screening due to low specificity and high false-positive rates. Spectroscopy-based techniques such as DRS, FS, and RS have various advantages compared to other optical methods. Specifically, the RS technique plays a vital role in retrieving molecular information from the breast tissue, which may help characterize various sub-types of breast cancer [58]. However, RS is not suitable for retrieving bulk optical properties of the breast tissue. Optical spectroscopy-based methods significantly help to understand the breast tissue's structural and functional information. The relative concentration of different constituents of the tissue-like total hemoglobin (HbT), collagen (C), lipid (L), and water (H₂O), as well as bulk properties of the tissue-like absorption (μ_a) and reduced scattering (μ_s') coefficient, oxygen saturation (SO₂) level, can be accurately quantified which play a vital role in thoroughly understanding of the metabolism of breast tissue. All these parameters can act as a basis for delineation between normal and cancerous tissues using spectroscopy-based techniques [53], [59], [60].

1) Optical Fiber-Based Diffuse Reflectance Spectroscopy (OF-DRS): In the diffuse reflectance spectroscopy (DRS) technique, the light source propagates through the breast tissue and results in the forward and backscattering of photons. The backscattered photons quantify the absorption

and scattering coefficients through the diffusion-based light propagation models [61], [62]. The DRS-based system consists of pairs of sources and detectors to quantify the mean optical tissue properties to analyze the morphology and composition of the tissue [63]–[65]. DRS measurements with a variety of photodetectors (i.e., spectrophotometers, photodiodes, charge-coupled devices (CCDs), etc.) can be helpful to get the optical properties of the tissue over a large field of view in minimal time [66]. Using fiber-optic cables (e.g., embedded in probe or biopsy needle), the DRS measurements can measure the optical properties over a small area or point measurement [67]. DRS with optic-fiber probe uses a pair of optic-fiber cables that are separated typically in the range of 1 mm to 8 mm. The analytical model using diffusion theory or analyzing the data sets with the help of measured intensity profile can be employed to analyze DRS measurement. The optical fiber-based sensors are compact, highly sensitive, biocompatible, and don't suffer from electromagnetic interference (EMI). These sensors can integrate with distributed sensing systems, making them a promising technology for point of care instrumentation [68]. Lu *et al.* [69] developed a hollow optical fiber-based probe integrated with the Fourier Transform Infrared Spectroscopy (FTIRS) system for *in-vivo* breast cancer diagnosis of 12 cancerous and six healthy tissues obtained from ($N = 12$) patients with 96% accuracy. Mohs *et al.* [70] reported a handheld 10 cm long Spectro Pen to detect malignant tumors. They used a tiny head unit for light transmission and collection employing fibers and a laser diode operating at 785 nm, and an indocyanine green (ICG) exogenous contrast agent to enhance the contrast and detect the tumor boundaries accurately in mice bearing 4T1 breast tumor, as shown in Fig. 3(a)–(c). Loyez *et al.* [71] reported a fiber-based surface plasmon resonance sensor (SPRS) for breast cancer diagnosis. This work used one cm-long fiber with a 400 μ m core diameter embedded with sputtered gold film for high sensitivity and reproducibility. Singh *et al.* [67] showed a portable, label-free, and compact multicore fiber-based sensor with high sensitivity for breast cancer and other cancer diagnoses. The seven cores of the reported sensor are arranged in a hexagonal pattern and immobilized with different nanoparticles like gold for high sensitivity, graphene oxide, and copper oxide for biocompatibility. The OF-DRS technique-based systems are non-destructive, easy to operate, may (or may not) require any dyes or chemicals as exogenous contrast, are affordable, and able to perform real-time analysis. OF-DRS-based systems possess high sensitivity, accurate quantitative assessment of tissues intraoperatively based on the fluorescence spectroscopy optical properties, and are easily integrated with the existing clinical equipment and tools with little or no primary safety consideration.

Co-registration between optical spectroscopy-based tools and histology is essential in transferring the novel optical spectroscopy system into clinical infrastructure. Still, it suffers from various critical limitations (e.g., tissue removal, micro-environmental heterogeneity, and deceased region are not easily accessible to the clinicians). An approach to address these limitations can be possible by integrating optical fiber with standard medical tissue-sampling equipment like core needle biopsy or

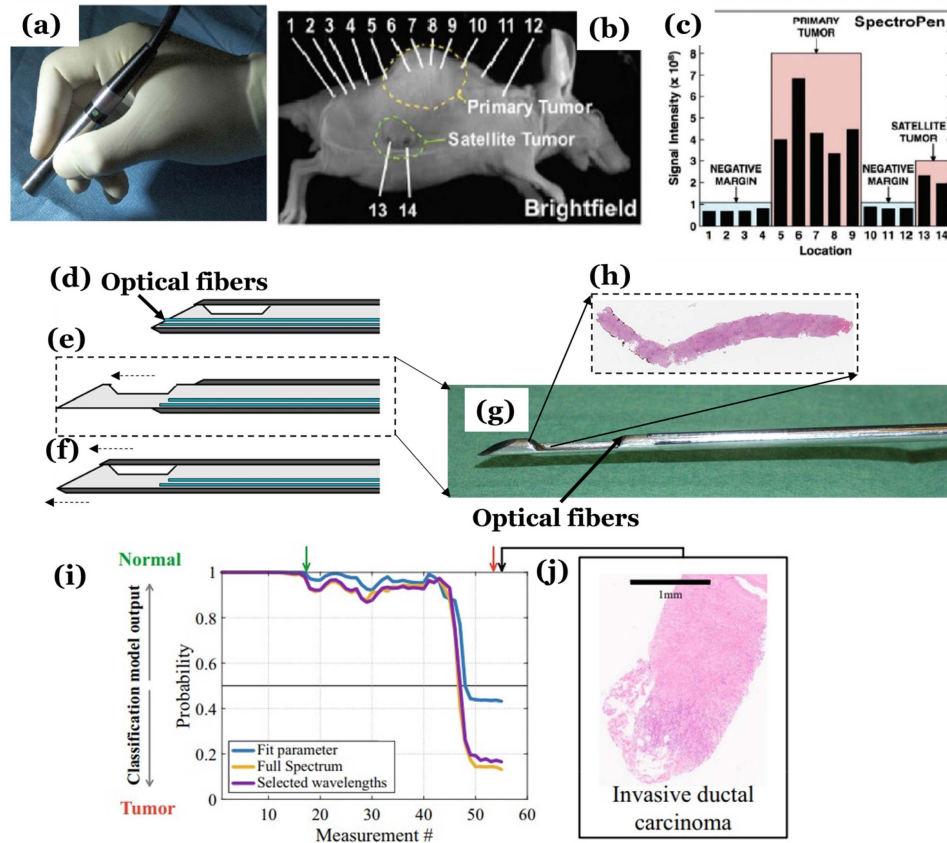


Fig. 3. Illustration of optical fiber based diffuse reflectance spectroscopy (OF-DRS) tools. (a) Actual image of Spectro Pen device held in clinician's hand which consists of illumination fiber (Ex), collection fiber (Coll), band-pass (BP) and long pass (LP) filter, mirror (M), and dichoric filter (D). (b) Brightfield image mentioning the anatomy and position of 4T1 breast and satellite tumor and numbers indicates the Spectro Pen measurement at these specific locations. (c) Spectro Pen recorded the signal intensities at different locations of the tumor [70], with the permission of ACS Publications, copyright 2010. (d) Biopsy needle integrated with optic fibers. (e) Cutting mechanism moves forward while optical fiber retract. (f) Outer stylet will move forward and cut the tissue in the periphery of biopsy cavity after pressing the release button. (g) Image with extended inner stylet. (h) H&E stained slide of the excised tissue obtained by the device. (i) Results obtained by the tool. (j) H&E image of excised tissue diagnosed with IDC [74], with the permission of Springer, copyright 2018.

biopsy forceps [72], [73]. De Boer *et al.* [74] developed an optical biopsy needle (i.e., biopsy needle integrated with optical fiber) based on the DRS method to diagnose breast cancer *in-vivo*, as shown in Fig. 3(d)–(g). The separation between lighting and collecting fiber is 1.36 mm with 1–2 mm penetration depth. It performs real-time breast cancer diagnosis based on fat/water ratio in cancerous and normal tissues obtained from ($N = 27$) subjects under the intraoperative guidance of ultrasound after processing the data using machine learning algorithms, as shown in Fig. 3(i). The classification model reported the mean accuracy (92%), specificity (89%), and sensitivity (94%), respectively, in the classification of adjacent normal and cancerous tissues. De Boer *et al.* [75] upgraded the OF-DRS-based system for surgical guidance during surgery, where a total of $N = 107$ breast specimens from $n = 107$ subjects diagnosed with ductal carcinoma in situ (DCIS) and invasive carcinoma (IC) using different classification models with varying sensitivity and specificity. Lam *et al.* [76] reported a narrow band (900–1000 nm) range OF-DRS system. The system consists of source fiber with a diameter of 3 mm and collection fiber with a diameter of 1 mm. The halogen lamp was used as a

source and spectrometer as a detector to quantify the fat/ water ratio in various phantoms and porcine tissue samples with high sensitivity.

2) Fluorescence Spectroscopy (FS): Fluorescence spectroscopy examines specific molecules' spectral properties (e.g., fluorophores) in tissue following illumination with a particular wavelength of light [77]. The concentration of endogenous fluorophores (i.e., collagen, lipid, elastin, tyrosine, and NADH) will determine the intensity and shape of the fluorescence spectrum in the target tissue. The differentiation between cancerous and normal tissue can be based on the molecule-specific fluorescence re-emittance properties [78]. Auto-fluorescence (AF) signal constitutes optical fingerprints of endogenous fluorophores, which play a vital role in tumor delineation [79]. J. E. Phipps *et al.* [80] developed a margin assessment system for breast cancer based on the fluorescence lifetime imaging (FLIm) technique that incorporates the support vector machine (SVM) classification algorithm that automatically detects the malignant and adjacent normal breast tissue. The experiments were performed on the excised breast tissues from ($N = 14$) patients and reported an accuracy of greater than 97% in detecting adipose, fibrous,

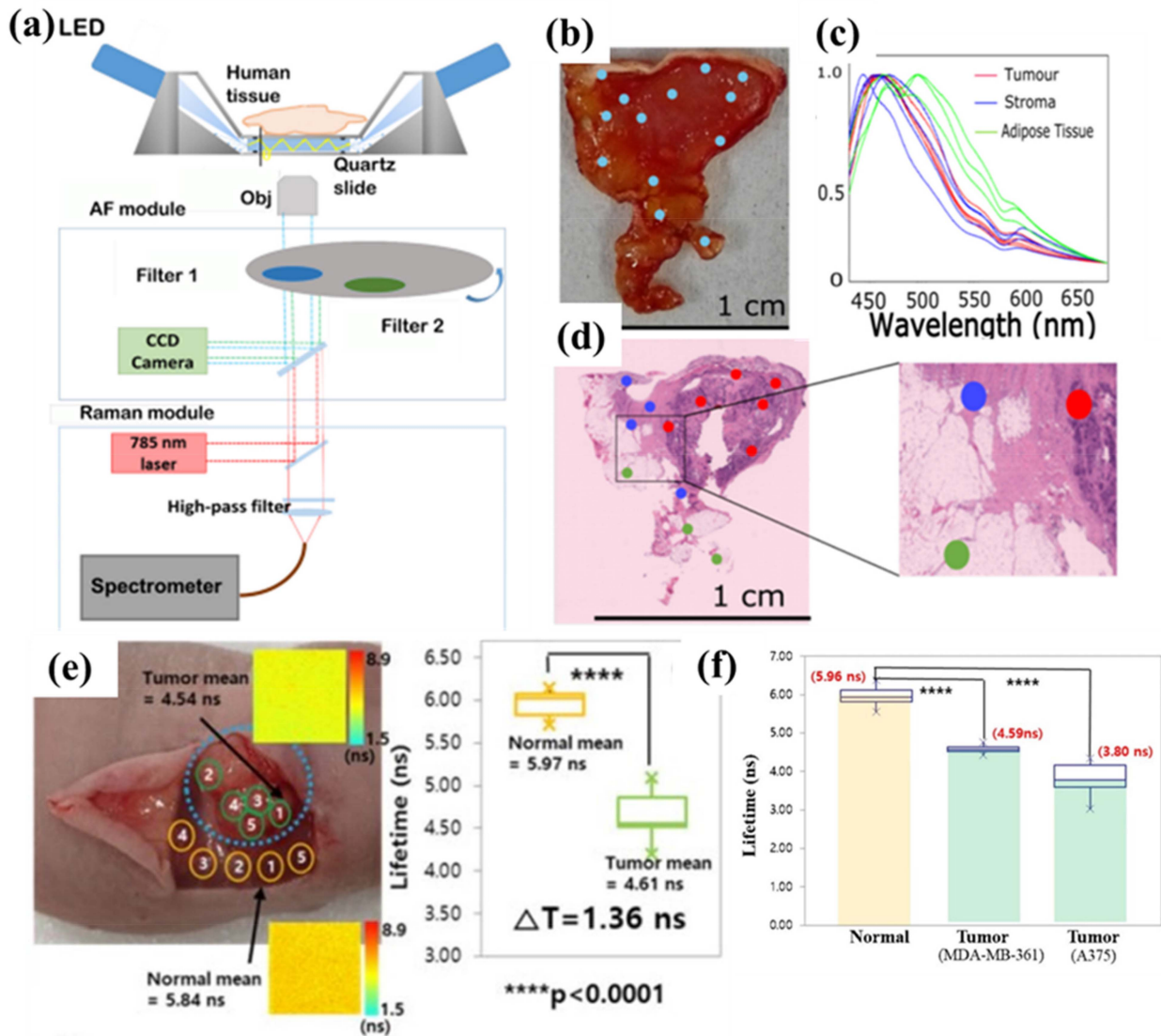


Fig. 4. Schematic of LED based TIR-FS-RS system. (a) Quartz slide on which tissue is kept and light emitted from the tissue is collected by objective lens. (b) Actual image of tissue sample. (c) Normalized fluorescence spectra with respect to the location specified by blue dots. (d) H&E image of the particular section of the tissue sample [86], with the permission of Biomedical Optics Express, copyright 2021. (e) Actual images depicting the point used for fluorescence lifetime measurement in mice and delineate the region between cancerous and normal region. (f) Difference between breast, skin tumor and normal tissue in fluorescence lifetime in mice [77], with the permission of Nature, copyright 2021.

and malignant tissues from the excised breast tissue specimens. Another study by J. Unger *et al.* [81] reported real-time cancer margin assessment using the label-free FLIm technique with a machine learning algorithm. In their research, total ($N = 18$) patients' lumpectomy and fresh mastectomy specimens were used and rapidly able to differentiate between cancerous and adjacent normal tissues with a specificity (93%) and sensitivity (89%). Lee *et al.* [77] designed an endoscopy probe using a pH-based fluorescence lifetime method for *in-vivo* breast cancer diagnosis in a rodent model that can be used as an alternative to the biopsy procedure, as shown in Fig. 4(e) and (f).

3) Raman Spectroscopy (RS): Raman spectroscopy technique relies on the inelastic scattering phenomenon in which the vibrational mode of the molecule (or fluorophore) changes due to the absorption of an incident photon on the tissue [82].

The alterations in the vibration mode of the molecule from one level to another trigger the generation of a photon with a shifted wavelength from the incident photon wavelength often called the Raman shift. The spectrum of RS describes the individual energy bands for particular molecular motions. It helps delineate cancerous and normal tissue because energy shifts are unique for every fluorophore [83]. NIR wavelength range of (785–840 nm) is suitable to obtain optimum Raman spectra due to minimal absorption, low auto-fluorescence, and high penetration depth inside the tissue [82]. RS is non-invasive, accurate, and gives real-time readings, but time-consuming and difficult to quantify depth-related information [84]. Auner *et al.* [85] extensively reviewed the theory behind RS and its clinical applications in breast, brain, prostate, and ovarian cancer. Lizio *et al.* [86] developed a multimodal technique using total internal reflection

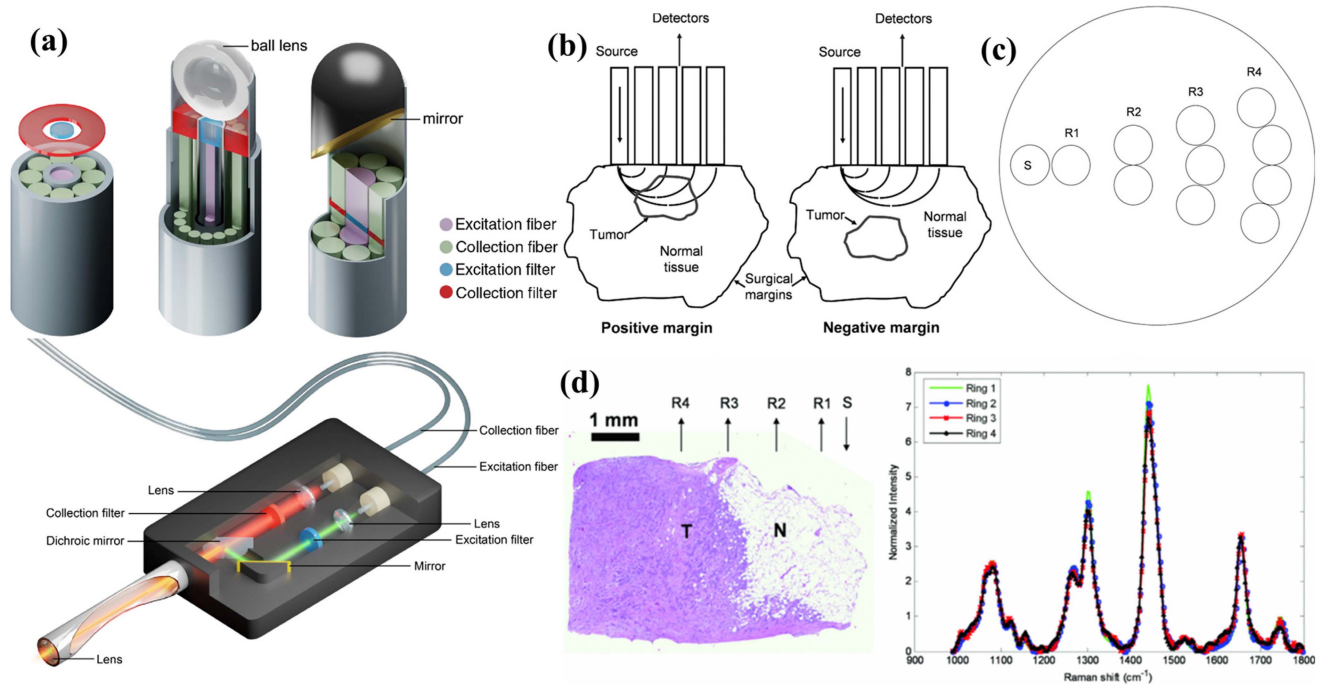


Fig. 5. Schematic of different Raman spectroscopy-based probes. (a) Endoscopic probe, with enhanced focus probe using ball lens, probe with provision of side-view, and handheld Raman spectroscopy based probe [84], with permission of SPIE, copyright 2018. (b) Probe based on the spatially offset Raman spectroscopy methodology designed by Keller *et al.* [89]. (c) Schematic of the probe with the placement of source and detector fibers separated by 1 mm. (d) H & E stained image and corresponding emission Raman spectra obtained from tumor and normal region of the breast tissue sample using SORS-based probe [89], with the permission of SPIE, copyright 2011.

(TIR), Auto-fluorescence (AF), and Raman Spectroscopy (RS) for rapid breast cancer margin assessment. This system used a LED operating at 785 nm for RS, whereas 365 nm was used for AF excitation, as shown in Fig. 4(a)–(d). The system took about 20 minutes to complete the tissue specimen analysis ($5.1 \times 7.6 \text{ cm}^2$) obtained from ($N = 35$) subjects and reported specificity and sensitivity of 80% and 90%, respectively. It was later reported that the background noise present in AF could be reduced significantly by choosing the appropriate operating wavelength [87]. Lazaro-Pacheco *et al.* [88] and E. Cardero *et al.* [84] comprehensively reviewed analytical and different optic fiber-based probe designs using RS, as shown in Fig. 5(a) for cancer diagnosis. Keller *et al.* [89] developed a probe based on the spatially offset RS (SORS) technique for intraoperative breast cancer margin assessment, as shown in Fig. 5(b) and (c). The group performed the *in-vitro* experiment using the probe on breast tissue samples ($N = 35$), reporting a sensitivity of 95% and specificity of 100%. The results obtained using the probe are shown in Fig. 5(d). M. Jermyn *et al.* [90] developed a label-free, LED-based multimodal (i.e., FS, RS, and DRS) spectroscopy probe for cancer diagnosis with a 100% sensitivity and 93% specificity. Kim *et al.* [91] extensively review optical fiber-based Raman spectroscopy, fluorescence spectroscopy, and multimodal (RS, FS, and DRS) optical spectroscopy systems for rapid cancer diagnosis *in-vivo*. FS has high SNR but suffers from low specificity, whereas, RS has high sensitivity and specificity but suffers from low SNR. Hence, a combination of FS and RS can mitigate the drawbacks of individual techniques and

provide detailed information on the metabolic and biochemical properties of the tissue [92].

A multimodal system combining RS and other optical modalities can detect breast malignancies in the tissue quickly, without sample preparation, non-destructively, and without using any external markers with improvements, resulting in translational diagnostic tool [92], as an adjunct to ultrasonography or mammography screening technologies in clinical settings in the future.

4) Photoacoustic Spectroscopy (PAS): Photoacoustic spectroscopy is a hybrid and fast-growing technology based on the photoacoustic effect. The microvascular networks, distribution of hemoglobin, and SO_2 levels are directly associated with the difference in the photoacoustic signals that can be mapped using two or more light sources of different wavelengths [93], [94]. The penetration depth and temporal resolution of the PAS modality are tunable in the diagnostic ultrasound transducer frequency range (1–10 MHz) and depend on multiple parameters such as the power of the light source, operating wavelength, and optical properties of background tissue. The recent development of PAS systems based on LED's [95], [96], and handheld micromachined ultrasound transducers [43] can lead to promising technologies for rapid breast cancer diagnosis.

Hariri *et al.* [97] developed an affordable and portable LED-based PAS system. The system comprises two LED arrays of 690 nm and 850 nm wavelength fixed on the sides of ultrasound transducers. The LED arrays can be tuned to different repetition

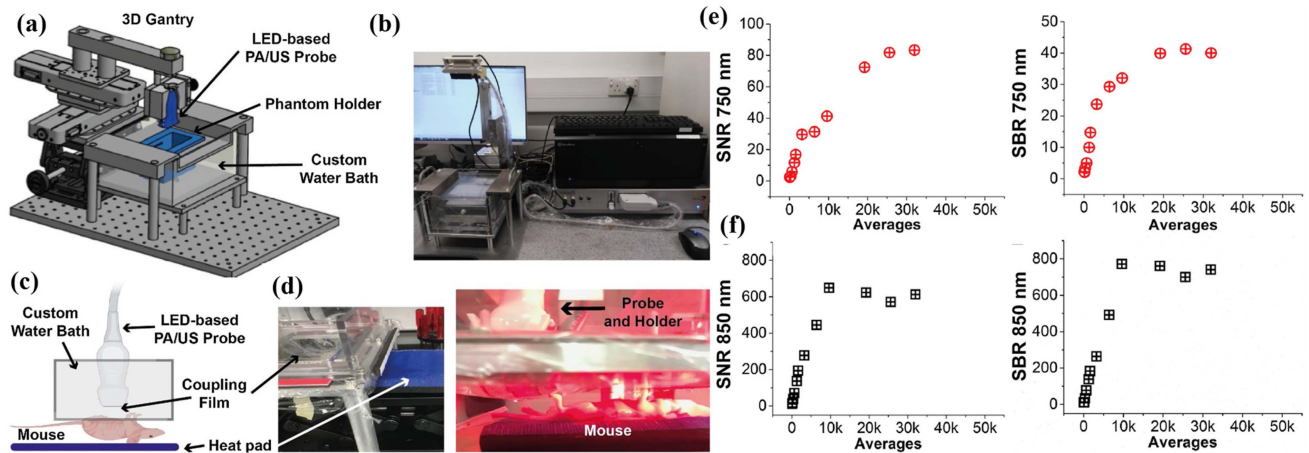


Fig. 6. (a) Schematic of LED-based photoacoustic system. (b) Actual images of the experimental setup used for phantom-based studies. (c) Schematic image of the LED-based handheld probe, custom water bath, and heat pad for animal-based studies. (d) Actual image of the probe performing studies on mouse and coupling and heat pad setup. (e) Signal-to-noise ratio (SNR) and signal-to-background ratio (SBR) at 750 nm. (f) SNR and SBR at 850 nm with error bars, respectively [98], with the permission of Elsevier, copyright 2021.

rates (e.g., 1 kHz – 4 kHz) with an axial and lateral resolution of 0.26 mm and 0.59 mm, respectively. Joseph *et al.* [98] reported a dual-wavelength (750 nm and 850 nm) LED-based system to perform temporal precision studies of the developed device after measurements on tissue phantoms and small animals over hours and days, as shown in Fig. 6(a)–(d). They reported high temporal stability in phantom and *in-vivo* studies and the impact of frame averaging on the estimation of signal to noise and signal to background ratios, as shown in Fig. 6(e) and (f). Balasundaram *et al.* [99] demonstrated the optoacoustic characterization on ($N = 14$) excised breast cancer specimens using eight different wavelengths ranging from (700–1100 nm) and delineating cancer from adjacent normal based on less or no lipid signals in the cancerous region compared to the adjacent normal region. Rodrigues *et al.* [100] developed a machine-learning (i.e., multi-class support vector machine (SVM) algorithm) based PAS system to assess the growth of breast cancer in a xenograft model with an accuracy of 95.2%. The proposed work is the first to use the combination of PAS and machine learning for breast cancer diagnosis with a high potential for clinical translation.

5) Polarization Sensitive Spectroscopy (PSS): Polarized light-based spectroscopy can play an essential role in understanding the heterogeneity or randomness of the tissue structure, which other optical spectroscopy techniques like diffuse optical spectroscopy are unable to quantify [101], [102]. The polarization-based study requires polarimeters like polarizers (i.e., linear or circular), quarter-wave plates (or retarders), depolarizers (or polarizer state analyzer (PSA)), and photodetectors. When a polarized light falls on the tissue samples, three polarization properties (i.e., depolarization, retardance, and diattenuation) can be quantified with the help of Stokes or Muller matrix parameters [102], [103] in different scattering scenarios [105]. The depolarization properties describe the tissue scattering phenomenon as tissue interacts with polarized light, resulting in loss of coherence within the tissue volume, leading to randomizing

the degree of polarization [104]. Retardance quantifies the phase change due to tissue constituents between its two orthogonal polarization states and yields birefringence [106], [107]. With the technological advancement in the field of LEDs and polymer-based thin polarizing film, a miniaturized polarization-sensitive system can be designed, as shown by Nishizawa *et al.* [108]. The group developed an endoscope device that consists of spin-LED and semiconductor-based spin-photodetectors based on the circular polarization technique to detect abnormal tissue from normal tissue *in-vivo* to quantify the anisotropy of the pseudo tissue using Monte Carlo simulation. Another recent work by Nishizawa *et al.* [109] reported cancer identification in a 40 μ m thick liver tissue sample using circular polarization and quantifying depolarization in the degree of circular polarization (DOCP) between cancerous and normal liver tissue samples. Kamal *et al.* [110] developed portable, LED-based polarization spectroscopy tools (i.e., POLS-NIRDx) to quantify the degree of linear polarization (DOLP) of human breast biopsy tissues *ex-vivo* at 850 nm and 940 nm wavelength, respectively. The developed system performed the experiment on excised tissue samples from ($N = 10$) patients and reported specificity (100%), sensitivity (100%), and accuracy (100%) for DOLP estimation at 850 nm, while DOLP estimation was reduced to specificity (70%), sensitivity (100%), and accuracy (80%) at 940 nm. These statistical parameters related to cancer and normal tissue delineation with high significance. Cancerous tissue has higher scattering due to structural change in the cells' nuclei, which can be quantified by the change in linear polarization of light scattered from the normal and cancerous tissue. Forward *et al.* [111] reported a flexible polarimetric probe using multimode optical fibers and thin polymer-based linear polarizing film to evaluate the muller matrix parameters and assess the biophysical properties of the chicken breast tissue *ex-vivo*. They quantify the mean retardance (103 ± 7 degrees) and mean depolarization ($44 \pm 7\%$) properties of chicken breast tissue and also reported the changes in the mean values of linear retardance (120 ± 9

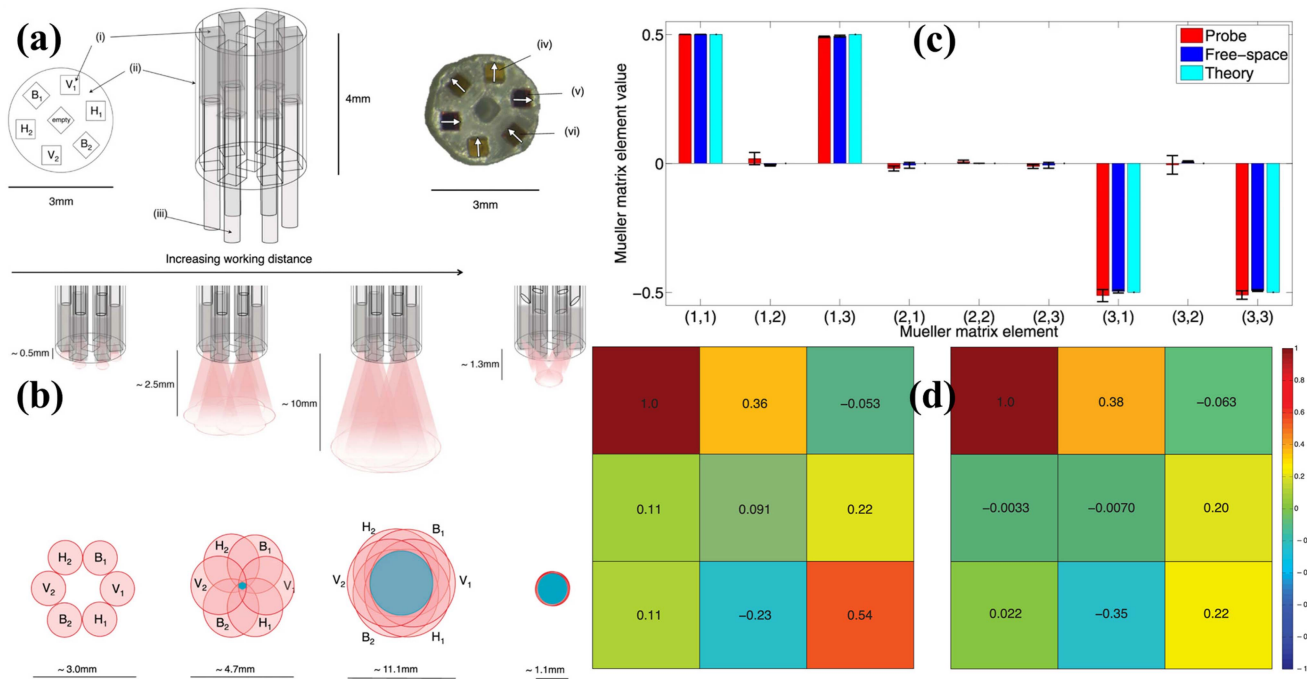


Fig. 7. Flexible polarimetric probe: (a) Top and side view of the flexible polarimetric probe where V, H, and B, corresponds to the different states of polarization. (b) Relationship between surface illumination profiles and working distance between probe and tissue. (c) Result obtained by the flexible polarimetric probe in terms of mueller matrix elements in three (i.e., probe, free-space, and theoretical values) different configuration. (d) Heat map for normal and stretched breast chicken tissue [111], with the permission of Nature, copyright 2017.

degrees) and depolarization ($50 \pm 12\%$) due to stretching of the chicken breast tissue, as shown in Fig. 7(a)–(d).

Since the 19th century, scientists have been studying the optical spectroscopy-based approach to diagnosing cancer. In the past few years, the technological advancements in the field of LEDs, computational methods (Artificial intelligence, machine learning, or deep learning-based software), and electronic circuits have enabled the design and development of affordable, robust, rapid, miniaturized, and highly sensitive devices which can be used in clinical practice.

B. Electrical Impedance Sensing

Bioimpedance measurement has been one of the most extensively researched techniques for malignant tumor detection because the electrical impedance of tissues is strongly dependent on their structure. The tissue's structure can be categorized into two fields or mediums: intracellular and extracellular. Herein, the intracellular medium constituted by the cellular membrane forms the resistive-capacitive part, and the extracellular medium, which mainly consists of water, electrolytes, and collagen fibers, forms a purely resistive part. Human tissues, therefore, are characterized by both resistive and capacitive behavior. Thus, tissue impedance is a complex quantity and has frequency-dependent transport as well.

Moreover, having a finite permittivity, the breast tissues can also be considered as a form of a dielectric. This frequency-dependent impedance is categorized into three zones of dispersions marked by three different frequency ranges. They are

α -dispersion (10 Hz–10 kHz), which indicates changes in the ionic environment around the cells, β -dispersion (10 kHz–10 MHz), which corresponds to cell structural relaxation, and γ -dispersion for higher frequencies, which is a signature from water molecules [112]. Of these, α -dispersion and β -dispersion are helpful as far as tissue impedance is concerned, as most of the effects due to carcinogenesis are reflected in these two ranges [113]. Recent advances have focused on needle-based impedance sensing devices, including hypodermic and biopsy needles. A significant advantage of such devices is that it provides a detailed assessment of the lesion in real-time, whether cancerous or normal, which is challenging when considering image-based techniques [114], [115].

1) Needle-Based Technologies for Electrical Impedance Measurement: In one of the first reports on needle-based impedance measurements, Mishra *et al.* [116] developed a probe by modifying a biopsy needle with an outer tissue cutting cannula and a shelf inside to hold the tissue. Polyimide (PI), a biocompatible polymer, was used to isolate the two coaxial parts of the needle, except the tip, to measure electrical impedance. The studies were performed by ultrasound-guided insertion of the modified biopsy needle into *ex-vivo* porcine and prostate cancer tissues, showing distinctly different behavior than the respective healthy tissues, as depicted in Fig. 8(a)–(c). In another study by Kang *et al.* [117], to overcome the challenge of transferring the pattern on the curved needle surface, the desired design was first created on a parylene-C film cast on a Silicon (Si) substrate. The film was then released post patterning to serve as a flexible photomask. This flexible photomask was then

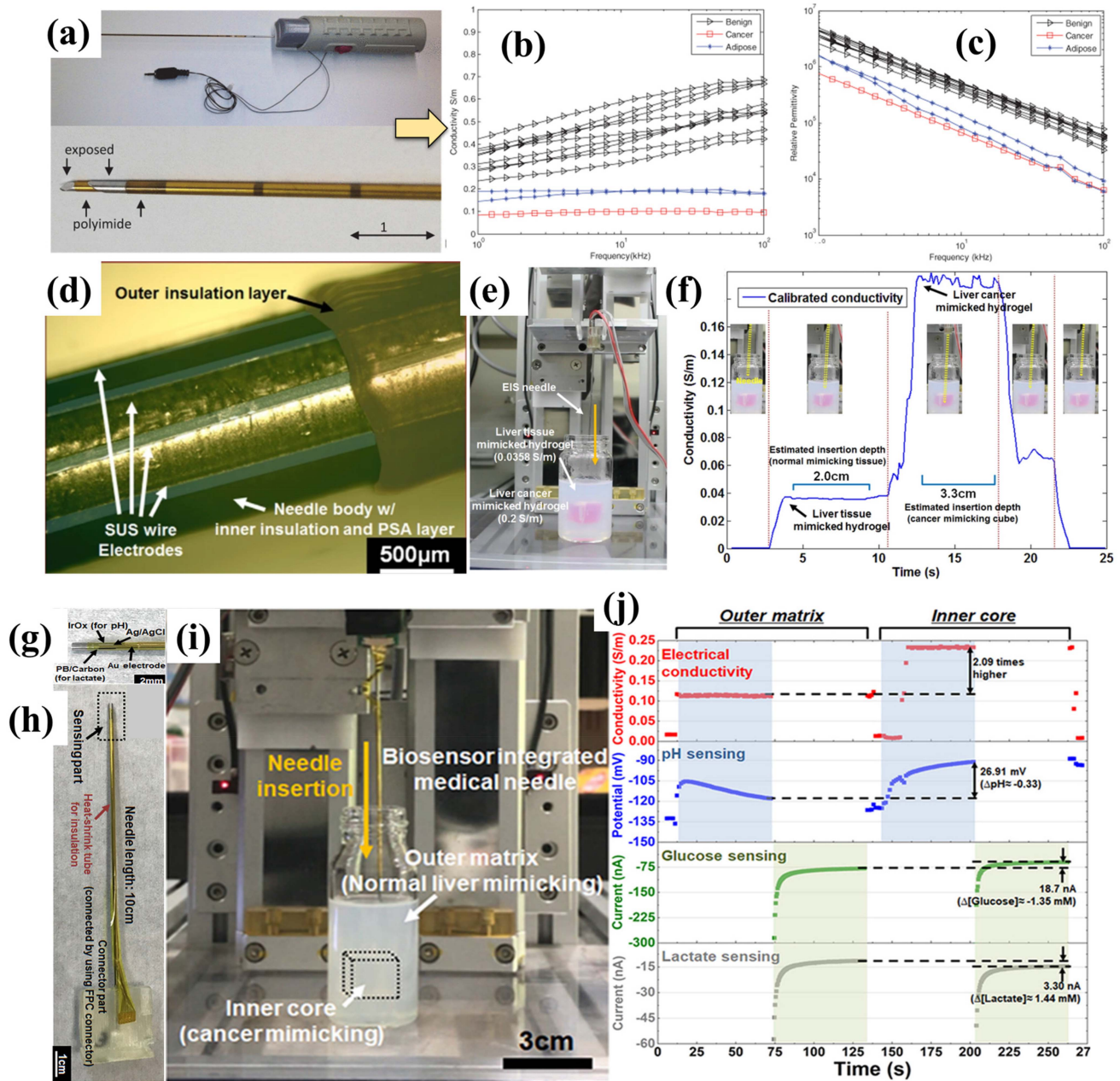


Fig. 8. Strategies for minimally invasive *in-vivo* electrical impedance measurement. (a) Biopsy needle based probe for electrical impedance spectroscopy, insulated with polyimide. (b) Conductivity. (c) Different value of relative permittivity measured by the device [116], with the permission of IEEE, ©2012 IEEE. (d) Four electrode patterns on a modified biopsy needle to minimize the electrode polarization effect. (e) Experimental setup for the measurements. (f) Calibration study performed on the hydrogel phantom [115], with the permission of Nature, copyright 2018. (g) Tip of the flexible multimodal sensor and constituting layers with their functionalities. (h) Actual image of multimodal sensor on the polyamide substrate and its cross-section along with FPC connector. (i) Experimental setup. (j) Characteristics of electrical conductivity, pH, glucose, and lactate sensing [120], with the permission of ACS, copyright 2020.

aligned to the photoresist spin-coated curved needle surface and then exposed to UV radiation, followed by metal deposition to obtain interdigitated electrodes (IDEs) on the needle surface [118], [119]. Another method to obtain micron-scale patterns on the curved surface of the biopsy needle is screen printing. This method offers a large-scale facile approach to patterning, albeit with a lesser resolution when compared to optical lithography. The recording electrodes can also be placed manually near

the tip region of the biopsy needle by using a biocompatible polymer that helps keep the electrodes glued to the polymer layer used to isolate the needle tip from the electrodes and employing the four-electrode method to reduce the EP effect, and obtained results, are shown in Fig. 8(d)–(f) [115]. However, the significant challenge here would be to place the electrodes, given the electrode placement uniformly is done manually. One approach that offers higher flexibility in design and processing

is to integrate a biopsy needle with a prefabricated sensor. The sensor needs to conform to the needle's curvature and be flexible enough to achieve this. Since flexible electronics have been researched extensively for many years, fabrication techniques have evolved considerably. This helps incorporate multiple functionalities in a single device without compromising the spatial resolution offered by micron-scale patterning. Recent work in this direction by Park *et al.* [120] demonstrated this flexible sensor's integration on the biopsy needle's hard surface, and obtained results are depicted in Fig. 8(g)–(j). The flexible probe was fabricated on a PI substrate spin-coated on a Si carrier wafer. After which, Ti/Au was deposited and then patterned using standard lithography techniques to define impedance recording electrodes. The as-prepared device was then integrated with a biopsy needle using a pressure-sensitive adhesive. The device could successfully help delineate normal tissue from cancer tissue in a porcine sample through measurement of variation in the impedance. In a recent work by Mahdavi *et al.* [121], a system termed an electrical pathological analyzer (EPA) was used for clinical evaluation of tumor margins of freshly excised breast tissues based on electrical impedance analysis. The EPA probe consists of a pair of hypodermic needles (inner diameter 1mm) modified to act as electrodes, interfacing with an impedance measuring unit. Based on impedance values at 1 kHz and impedance phase slopes, the EPA system could sense the location of malignant cells in tumor margins until 3 mm depth.

C. Mechanical Sensing

The Mechanical sensing technique helps understand the disease progression in the biological tissue and planning any surgical operation (e.g., breast implants or plastic surgery) [122]. The lobes of the breast that produce milk and the ducts that transport milk to the nipple are made up of fibrous and glandular structures. Fibrous and glandular tissue are combined to form fibro glandular tissue, and the space between them is filled with fatty tissues [123]. Mechanical properties of the breast tissues (i.e., fat, fibro glandular, and tumor) are highly dependent on their tissue microenvironment. As the disease progress, a lot of physiological changes occur in the tissue microenvironment. Quantifying these changes with the help of different mechanical sensing technologies (such as MEMS-based sensors, optical coherence micro elastography (OCE), or ultrasound-based systems) can help in breast cancer diagnosis at an early stage [124]–[127]. During a women's life cycle, the tissue distribution inside the breast undergoes different cyclic changes, and these changes depend upon age, hormones, pregnancy, and the menstrual cycle. All these changes show a profound effect on the tissue morphology and composition. Cancerous tissues are stiffer than normal and show high non-linear stiffening at high deformation [128]. As a result, precise measurement of tissue mechanical properties such as elasticity and viscoelasticity can aid in breast cancer diagnosis. Conventional techniques for characterizing mechanical properties use indentation probes or compression techniques [129], [130]. However, traditional methods cannot support *in-vivo* analysis. Thereby, understanding the mechanical properties of breast tissue using MEMS-based structure and

integration with machine learning [131] based computational methods facilitates a minimally invasive technique for quantifying the mechanical properties of breast tissue.

1) Mechanical Sensing Using MEMS-Based Devices for Breast Cancer Diagnosis: The elastic/stiffness properties of malignant and normal human breast tissue have been studied extensively in many studies [122], [130]–[132]. MEMS comprises electrical and mechanical elements ranging from a few micrometers to hundreds of micrometers. Actuators and sensors are two broad categories for MEMS-based devices. However, few studies have employed MEMS-based sensors to quantify the mechanical properties of the breast cells at micro or nanoscale levels [134], [135]. Although MEMS-based sensors can be integrated into portable lab-on-chip systems, thus it can be a viable option for a portable diagnostic system. However, MEMS dice must interact with the surrounding region of interest for proper sensing and signaling, so particular care must be taken when packaging them [136], [137]. In this direction, Park *et al.* [138] reported a flexible MEMS-based biochip consisting of a microheater and piezoresistive force sensor for breast cancer diagnosis using electromechanical coupling factor as a new biomarker for breast cancer diagnosis, as shown in Fig. 9(a) and (b) and obtained results in Fig. 9(c) and (d). Pandya *et al.* [125] developed a compact portable tool integrated with a MEMS-based disposable biochip to quantify the mechanical (i.e., stiffness) and electrothermal (i.e., electrical resistivity and thermal conductivity) properties of deparaffinized breast tissue. The reported tool can differentiate between cancerous and normal tissue and provide information about disease progression in tissues. Another work by Pandya *et al.* [139] reported a sensor array on a flexible substrate for characterizing breast cancer and quantified the mechanical (i.e., strain) as well as electrical (i.e., resistivity) properties of formalin-fixed paraffin-embedded breast tissues to successfully differentiate between malignant and benign region.

2) Contact Force Sensing: Contact force sensing is another mechanical sensing technique that can effectively characterize and quantify the malignancy in tissue through a MEMS-based, highly sensitive micro-force sensor [140]. Atomic force microscopy (AFM) is a conventional technique to quantify the mechanical properties of a biological tissue sample with the help of a piezoelectric material-based microcantilever structure [141]. While probing the tissue, the deflection in the microcantilever is optically sensed and quantified sample stiffness [142], [143]. However, the AFM system consists of complex electronics and optical assembly, bulky, and incompetence to analyze opaque liquids. Commercially available AFMs typically consist of a single cantilever probe which is very time-consuming in analyzing histopathological studies of the tissue. A piezoresistive or piezoelectric-based sensing mechanism can be a better alternative to characterize the tissue's mechanical properties without the requirements of complex readout electronics. The working principle of the piezoresistive transducers is to translate the force acting on it into a resistance value. It is well-known that single-crystal silicon has a strong piezoresistive effect [144]–[149]. In this direction, Pandya *et al.* [140] reported a piezoresistive microcantilever force sensor to characterize the

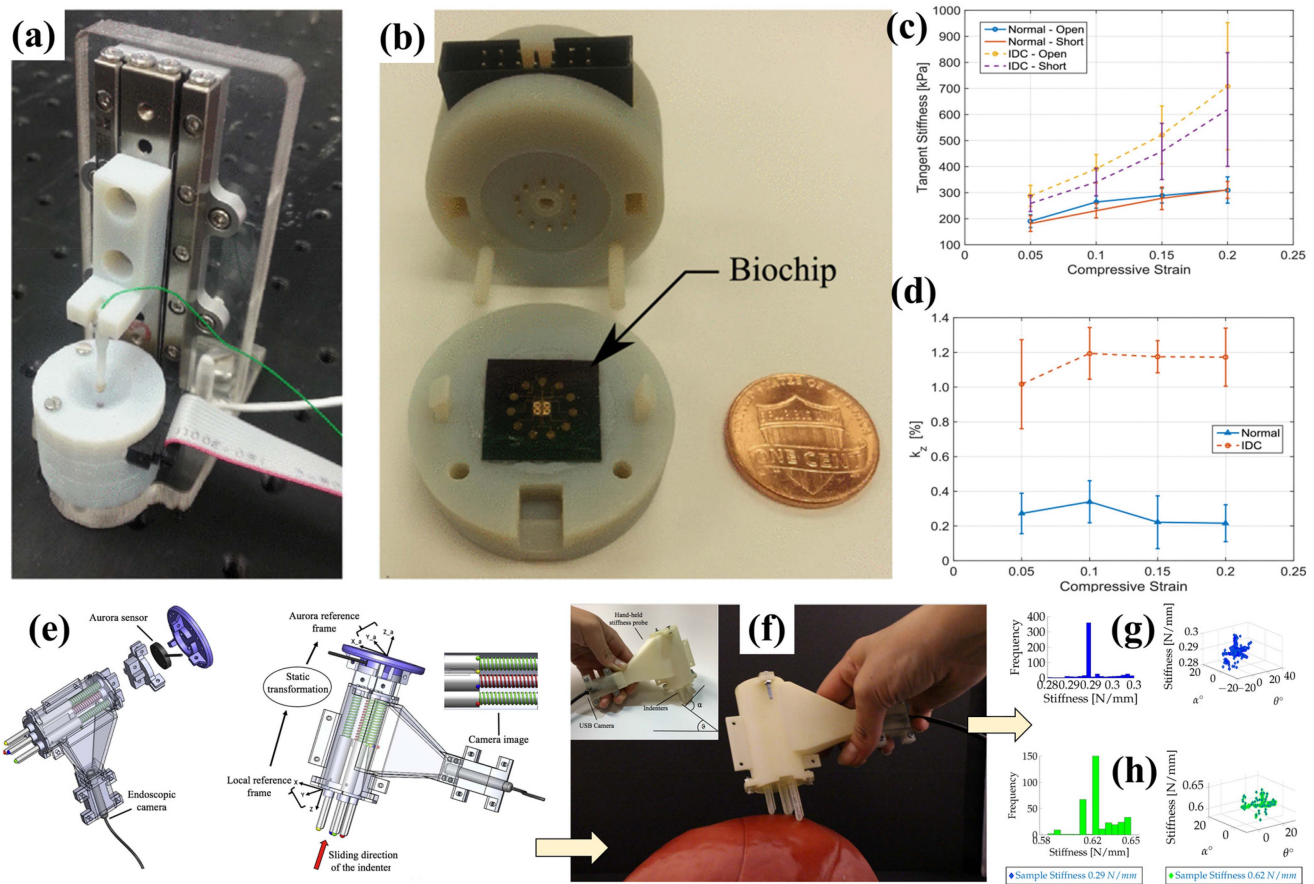


Fig. 9. MEMS-based sensors for mechanical characterization of breast tissue. (a) Experimental setup of the diagnostic tool. (b) Fabricated MEMS-based biochip integrated with the package. (c) Experimental result in terms of tangent stiffness. (d) Electromechanical coupling factor of the breast tissue [138], with the permission of IEEE, copyright IEEE, 2017. (e) Schematic of exploded view and working principle of the stiffness probe. (f) Actual image of handheld stiffness sensor with multiple axis performing tissue characterization. (g) Result obtained in terms of stiffness with respect to orientation angle of the hand-held probe. (h) Variations in the stiffness values varying with respect to pan and tilt angle [143], with the permission of MDPI, copyright 2018.

benign and malignant tissues of the breast. The reported sensor is ($260 \mu\text{m} \times 35 \mu\text{m} \times 2 \mu\text{m}$) with a resistivity of order ($1.316 \times 10^{-3} \Omega \text{ cm}$), fabricated using standard silicon micromachining technology [150]. A cylindrical SU-8 tip with a thickness of $8 \mu\text{m}$ was used to indent the tissue, and observed the change in resistance was higher for benign breast epithelial tissue than cancerous tissue cores, which supports that tissue stiffer as disease progress [151]. Faragasso *et al.* [143] reported a handheld stiffness probe to quantify the mechanical properties of soft tissue during palpation, as shown in Fig. 9(e) and (f), and obtained results in Fig. 9(g) and (h). Somashekhar *et al.* [152] reported a handheld and portable point-of-care medical imaging device named iBreast Exam to detect breast lesions at early stages for younger and older women. This handheld device consists of a 4×4 array of pressure sensors made up of PZT integrated with a tablet, having specificity and sensitivity of 93% and 85%, respectively. Egorov *et al.* [153] developed an affordable, portable system that is easy to operate for mechanical imaging of breast called SureTouch. The reported system contains a probe with a force sensor array and a data acquisition unit that directly connects to the laptop through a USB port and visualizes the pressure

pattern as the force is applied on the breast. Pandya *et al.* [154] reported a piezoresistive MEMS-based indenter to characterize the cancerous and normal region of breast tissues and produce a better delineation between normal and diseased regions by converting the indented force to change in resistance values to the corresponding voltages values. However, this technique is difficult to describe the chemical and biophysical properties of cancerous tissue. Table I (Supplementary files) summarizes the different optical, electrical, and mechanical sensing technologies based tools for breast cancer diagnosis.

V. RECENT ADVANCES IN MULTIMODAL SENSING TECHNIQUES

Multimodal sensing techniques are promising technology to rapidly diagnose cancers with high efficacy and accuracy. Pandya *et al.* [125] reported a portable tool to quantify the electrical-thermal-mechanical properties of breast tissue by using a disposable MEMS-based biochip. They reported significant differences in electrical resistivity, thermal conductivity, and mechanical stiffness properties of the cancerous and

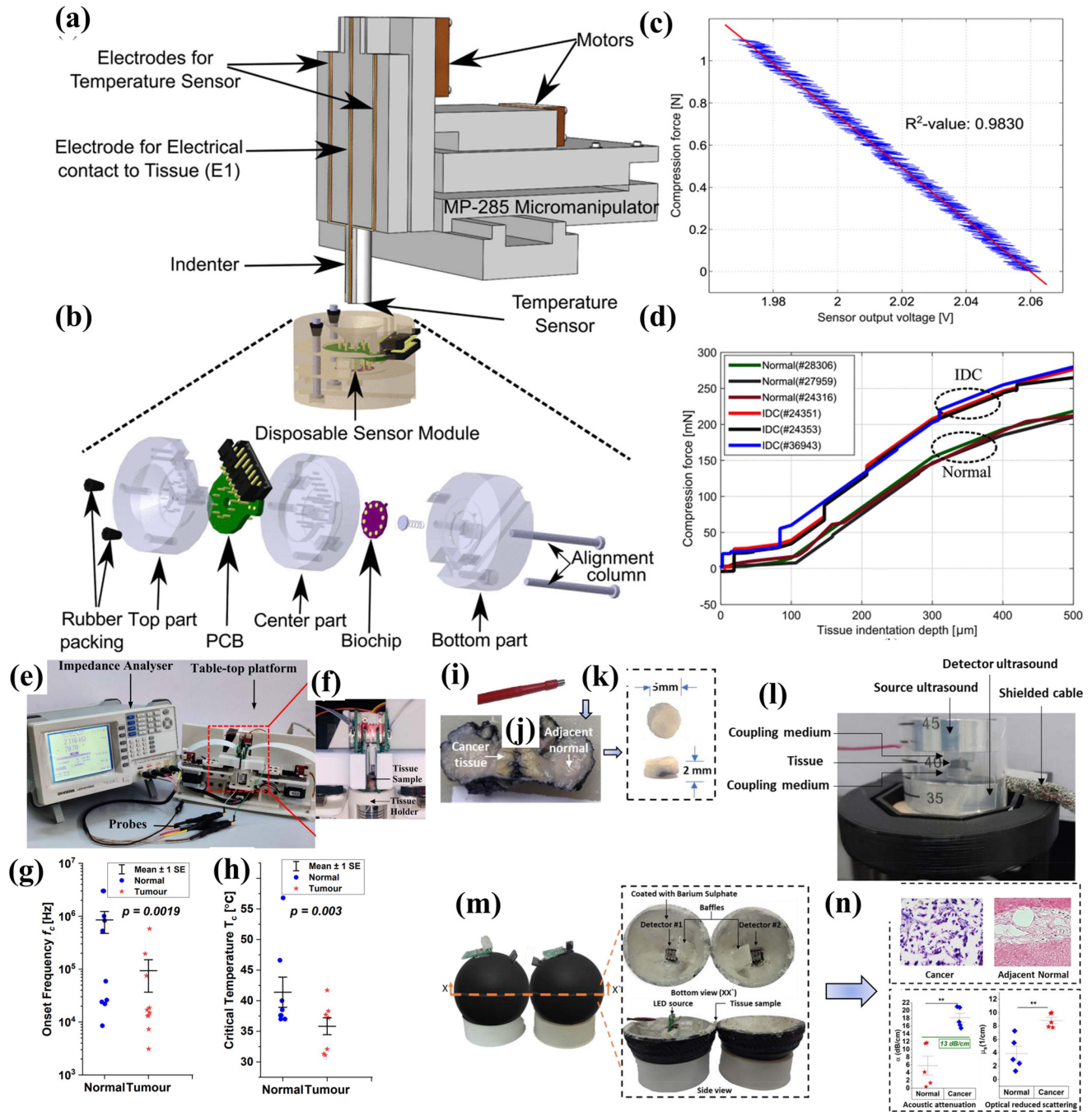


Fig. 10. Multimodal system for breast cancer diagnosis. (a) Schematic of the portable multimodal system. (b) Exploded view of the MEMS-based sensor array along with packaging system, and results obtained by the system. (c) Calibration curve of the force sensor. (d) Force curve acquired after indenting breast tissues [125], with the permission of IEEE, copyright IEEE, 2016. (e) Bimodal system for characterizing electrical and thermal properties of breast tissue. (f) Tissue sample sandwiched between the MEMS-based sensor and results obtained by the system. (g) Onset-frequency. (h) Critical temperature of tumor and adjacent normal tissue [155], with the permission of IEEE, copyright IEEE, 2021. Optical and acoustic modality based multimodal system. (i) Punch biopsy. (j) Excised malignant tissue along with normal tissue. (k) Core of the breast tissue used for measurements. (l) Module for acoustic measurements. (m) Module for optical measurements. (n) Result obtained by the hybrid spectral-IRDx system [156], with the permission of IEEE, copyright IEEE, 2021.

adjacent normal breast tissue and could be used as potential biomarkers for breast cancer diagnosis, as shown in Fig. 10(a)–(d). Vishnu *et al.* [155] reported a portable and affordable bimodal tool to quantify the on-set frequency and electrical transport in cancerous and adjacent normal tissue to diagnose breast cancer, as shown in Fig. 10(e)–(h). Pal *et al.*

[156] reported a portable hybrid spectral-IRDx multimodal system based on the optical and acoustic modality for breast cancer diagnosis. They reported reduced scattering coefficient and acoustic attenuation coefficient could be used as a biomarker for delineation between malignant and adjacent normal breast tissue, as shown in Fig. 10(i)–(n). Optical fiber-based

multimodal biophotonic technologies can be suitable for breast cancer margin analysis by utilizing conventional endoscopes in *in-vivo* situations. In this direction, Kim *et al.* [58] extensively reviewed the possible combinations of different optical modalities-based techniques for different types of cancer diagnoses. Overall, a multimodal system incorporating optical, electrical, and mechanical modalities has a great scope for breast cancer diagnosis and margin assessment in intraoperative settings. Table II (Supplementary files) summarizes the different multimodal-based systems used for breast cancer diagnosis.

VI. UNMET CHALLENGES IN OPTICAL, ELECTRICAL, AND MECHANICAL SENSING TECHNIQUES

One of the critical challenges for optical spectroscopy as a diagnostic tool is the unavailability of medically compatible miniaturized probe designs that are easy to operate without frequent calibration. The calibration methods are currently based on power meters, tissue phantoms, spectrophotometry, detection methodologies, and reflectance standards before or after measurements. Various limitations associated with these calibration techniques like instrument fluctuations (e.g., wavelength drift, fiber bending losses, and impedance mismatch) in real-time, extensive setup time, and slow data processing [157]. Ultimately, an affordable, portable, rapid, robust, and accurate system would be preferred for intraoperative settings. Especially in breast cancer diagnosis, a handheld optical spectroscopy-based system is considered a promising technique that can aid the existing gold standard technique (i.e., histopathology) to reach an informed decision quickly. Translating an innovative technology into clinical practice is challenging and requires significant investments, time, large-scale trials, and various regulation issues [158]. The technical advancement in microfabrication technologies (e.g., laser micro-machining and focused ion beam sputtering), light sources (e.g., high power multiwavelength LEDs), low noise photodiodes, and application-specific integrated circuits (ASICs) can make it possible and hold great potential for the development of a compact and novel diagnostic optical spectroscopy-based tools for breast cancer diagnosis [159]. In addition, artificial intelligence-based data processing algorithms (e.g., artificial neural networks, genetic algorithms, support vector machines, etc.) can play an essential role in accurate and rapid data processing in real-time [91]. However, machine learning-based techniques have challenges, such as limited training dataset range, incorrect decision making due to biases, overfitting, etc. Finally, organ-specific optical spectroscopy devices can significantly improve the efficacy of breast cancer diagnosis. Finally, the sterilization of the optical modality-based probe used recurrently in clinical setting is also a challenge; hence newer non-contact probes are required to be designed.

Although electrical impedance sensing has been one of the methods for breast tissue delineation, this technique has some inherent limitations. Electrical impedance can only provide information about the tissue at the surface, even when measured using biopsy needle-based probes [160]. Applications such as identifying tumor margins, surface profile, and the depth profile

of the tissues are also required for an accurate assessment. Also, post-processing of the information obtained from electrical impedance sensing is an evolving field that reduces the false positive rate [161], [162]. In this regard, machine learning algorithms have improved the classification accuracy of breast tissues. In addition, measurement artifacts caused due to electrode polarization effects and constantly changing TME, for example, in *in-vivo* scenarios, are some of the other challenges that need to be addressed. Using an additional modality such as pH, reactive oxygen species (ROS) sensing, along with electrical impedance can be a practical approach in the future to improve the sensor's ability to delineate normal from the cancerous region in *in-vivo* scenarios, as explored in one of the recent works [114]. Patient-to-patient variability in electrical impedance value when considering similar types of breast tissues is another challenge encountered when electrical impedance sensing is used as a standalone modality [163], [164]. Integrating additional modalities such as pH and ROS sensing to electrical impedance sensing can help address this issue by providing real-time information about the TME and thus help correctly identify tumor margins for a large cohort.

Most of the literature reports on mechanical sensing for breast tumor diagnosis have focused on *ex-vivo* tissues. However, the properties of tissues change significantly when excised and preserved in formalin or paraffinized. It is desirable to develop sensors that can perform mechanical characterization of the tissues in an *in-vivo* scenario to improve tumor margin assessment. This is one of the significant challenges in mechanical sensing; wherein there is a need to develop a miniaturized probe design suitable for *in-vivo* tissue characterization. This would require efforts to develop novel fabrication techniques while considering the biocompatibility of materials used for sensor fabrication, residual stress compensation techniques during fabrication of sensors, and data acquisition & processing units for real-time analysis.

VII. CONCLUSION

The multimodal physical sensing-based device offers a distinct perspective on breast cancer diagnosis. These hybrid technologies combine the existing ubiquitous surgical tools like biopsy needles with flexible MEMS-based device fabrication, which have shown exciting results in *ex-vivo* and *in-vivo* situations. These devices have quickly evolved from being used for measuring a single bulk tissue properties to multiple properties such as electrical impedance, optical absorption, and scattering coefficients, the concentration of tissue chromophores (e.g., lipid, collagen, and total hemoglobin), and tissue elasticity/stiffness providing more reliable and accurate information about tumor microenvironment in the event of breast cancer. The most frequent unmet challenges these multimodal systems face are unable to respond effectively to tissue microenvironment, the effect of foreign body response on the functionality of the sensor performance in implantable scenarios, biocompatibility, flexibility, frequent calibration, and data acquisition and processing. This 'hybrid' approach of device integration can be further extended to newer bio-sensing modalities like

pH, Reactive Oxygen Species (ROS), high-quality, in-depth imaging, intraoperative breast margin assessment, and targeted drug delivery. Advances in material science, especially nanomaterials and nano-structuring, have greatly aided this accelerated development and will continue to do so in the future. The future directions would be to convert these physical sensing modalities into quasi-implantable devices to fully implantable devices and incorporate novel ideas to integrate optical, electrical, mechanical, and chemical sensing modalities after mitigating the discussed unmet challenges faced by each modality. Wireless control of such devices could be the next challenge in this field which would require extensive research on power management and transmission routes ranging from electromagnetic to optical. In this review, the recent reports on integrating the clinical tools are discussed and analyzed along with the current unmet needs, which could address the challenges of breast cancer diagnosis.

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